

Project Documentation: Kubernetes Application Deployment Automation using KEDA

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Repository: [Assignment Simplismart \(Public\)](#)

1. Overview

This project automates Kubernetes cluster operations using a Bash CLI script (simplismart.sh) and a Jenkins pipeline (Jenkinsfile).

Key features:

- Check connectivity to a Kubernetes cluster.
- Installs Helm and KEDA (Kubernetes Event-Driven Autoscaling).
- Deploys applications with auto-scaling via KEDA.
- Provides health monitoring for deployments.

2. Prerequisites

- Kubernetes cluster: kubectl configured.
- Jenkins Server: kubectl and helm binaries available in the agent's \$PATH.
- Git Repository: GitHub to host the project (for Jenkins integration).
- Git (to clone the repository).

3. Script Usage

CLI usage (simplismart.sh)

Execute with the following commands:

- 1) To install the Helm and keda please run the below command .

```
./simplismart.sh install
```

Description: This command automates the setup of Helm and KEDA on a Kubernetes cluster, ensuring all dependencies are configured for event-driven autoscaling. It validates cluster access and deploys KEDA with Helm

- 2) To Create the application please run below command.

```
./simplismart.sh -n <namespace> create
```

Description: This command deploys an application with KEDA autoscaling in the specified Kubernetes namespace, automatically creating deployments, services, and scaling rules.

- 3) To check the health status of the application please run below command.

```
./simplismart.sh -n <namespace> status
```

Description: This command checks the health and metrics of deployments in a specified Kubernetes namespace, showing pod status, resource usage, and recent events.

4. Jenkins Pipeline (Jenkinsfile)

1. Pipeline Setup

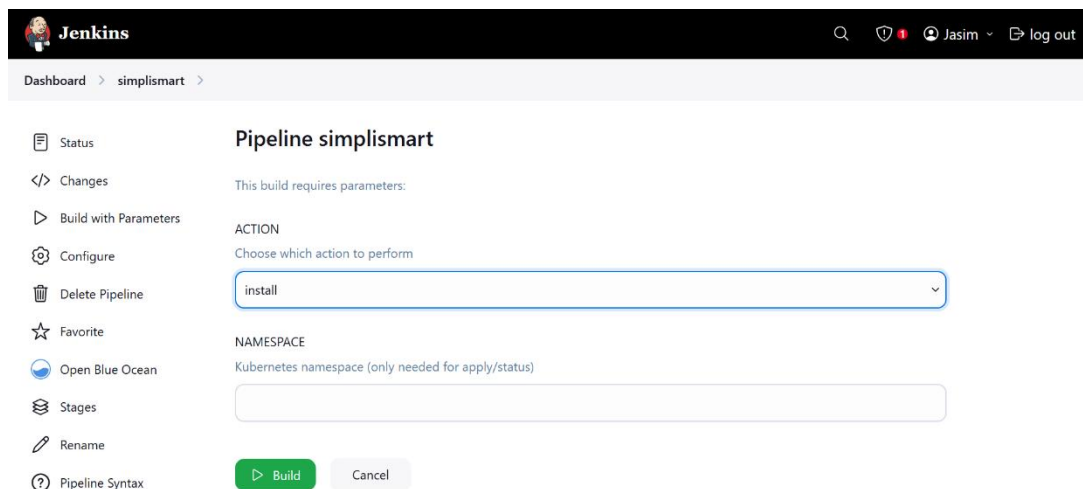
1. Create New Jenkins Job:
 - Select Pipeline
2. Configure Pipeline Script:
 - Set Pipeline Script from SCM (Git repo URL + branch)

2. Execution Steps

1. Manual Trigger:
 - Navigate to job → Build with Parameters
 - Select action: install/create/status
 - For create/status : Specify namespace

3. Jenkins Screenshots

a. Install



The screenshot shows the Jenkins web interface. At the top, there's a header with the Jenkins logo, a search icon, a shield icon, a user profile icon labeled 'Jasim', and a 'log out' link. Below the header is a breadcrumb trail: 'Dashboard > simplismart >'. On the left side, there's a sidebar menu with icons and labels for 'Status', 'Changes', 'Build with Parameters', 'Configure', 'Delete Pipeline', 'Favorite', 'Open Blue Ocean', 'Stages', 'Rename', and 'Pipeline Syntax'. The main content area is titled 'Pipeline simplismart'. It contains a message 'This build requires parameters:' followed by an 'ACTION' section with the text 'Choose which action to perform'. Below this is a dropdown menu with 'install' selected. Underneath is a 'NAMESPACE' section with the text 'Kubernetes namespace (only needed for apply/status)' and an empty text input field. At the bottom of the configuration area, there are two buttons: a green 'Build' button with a play icon and a grey 'Cancel' button.

b. Create

Jenkins

Dashboard > simplismart >

Status

Changes

Build with Parameters

Configure

Delete Pipeline

Favorite

Open Blue Ocean

Stages

Rename

Pipeline Syntax

Pipeline simplismart

This build requires parameters:

ACTION

Choose which action to perform

create

NAMESPACE

Kubernetes namespace (only needed for apply/status)

simplismart

Build Cancel

c. Status

Jenkins

Dashboard > simplismart >

Status

Changes

Build with Parameters

Configure

Delete Pipeline

Favorite

Open Blue Ocean

Stages

Rename

Pipeline Syntax

Pipeline simplismart

This build requires parameters:

ACTION

Choose which action to perform

status

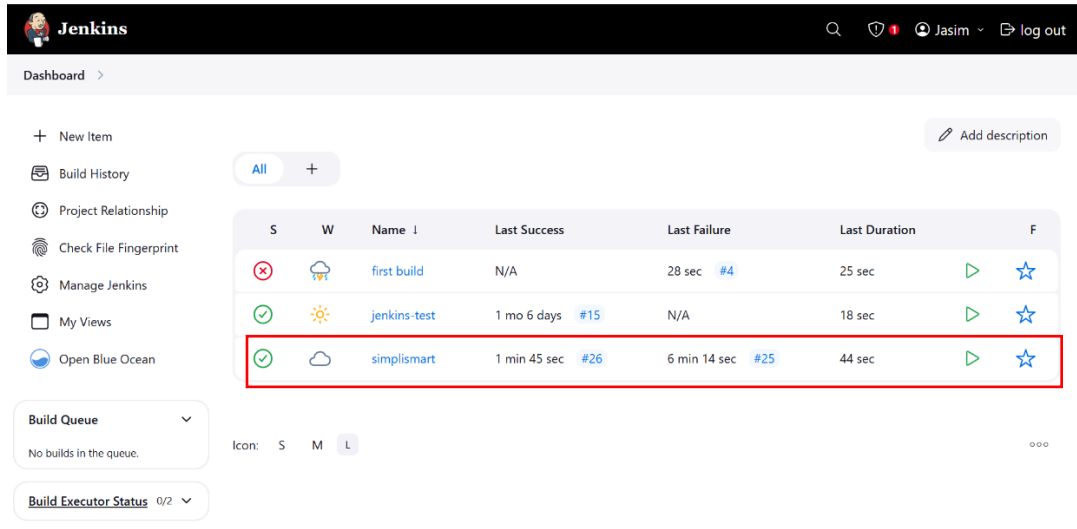
NAMESPACE

Kubernetes namespace (only needed for apply/status)

simplismart

Build Cancel

d. Post Pipeline execution



The screenshot shows the Jenkins Dashboard with a sidebar on the left containing links like 'New Item', 'Build History', 'Project Relationship', 'Check File Fingerprint', 'Manage Jenkins', 'My Views', and 'Open Blue Ocean'. The main area displays a table of build history. The table has columns for 'S' (Success), 'W' (Warning), 'Name', 'Last Success', 'Last Failure', 'Last Duration', and 'F' (Favorite). The 'simplismart' build is highlighted with a red box. Below the table, there are sections for 'Build Queue' (No builds in the queue) and 'Build Executor Status' (0/2).

S	W	Name	Last Success	Last Failure	Last Duration	F
✗	☁	first build	N/A	28 sec #4	25 sec	▶ ☆
✓	☀	jenkins-test	1 mo 6 days #15	N/A	18 sec	▶ ☆
✓	☁	simplismart	1 min 45 sec #26	6 min 14 sec #25	44 sec	▶ ☆

5. Output Logs

CLI

Script execution logs via CLI are pushed to Git repo and can be accessed [script_execution_logs_cli.txt](#).

Jenkins

Script execution logs via jenkins are pushed to Git repo and can be accessed [script_execution_logs_jenkins](#).

6. Application Validation

Once script is executed via CLI or Jenkins, please open the browser and go to <http://localhost:30080> as service is being exposed at 30080 static port to outside world.

Post clicking url you will be able to see banner message as shown below :

